Gravitational effects on oscillon lifetimes

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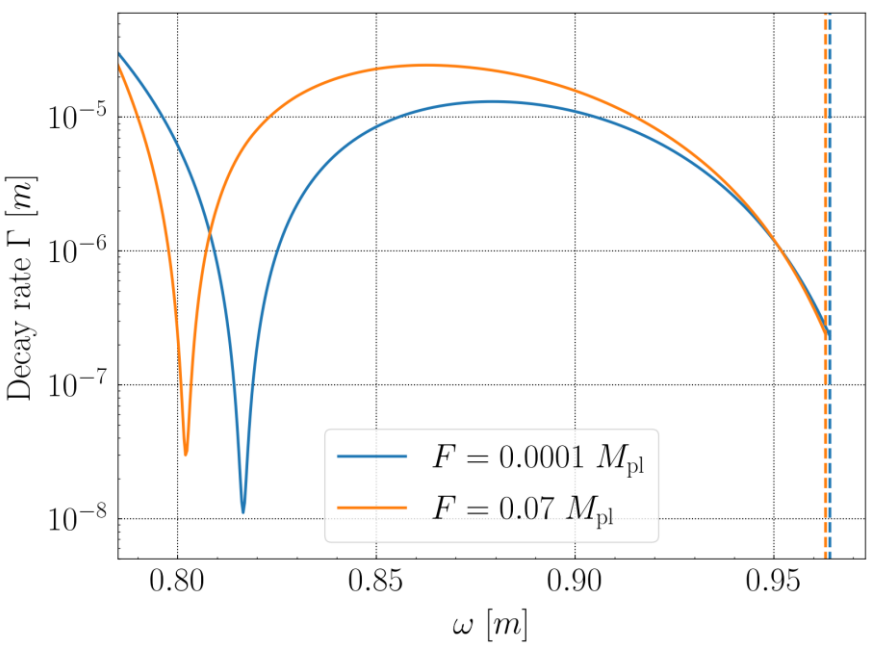
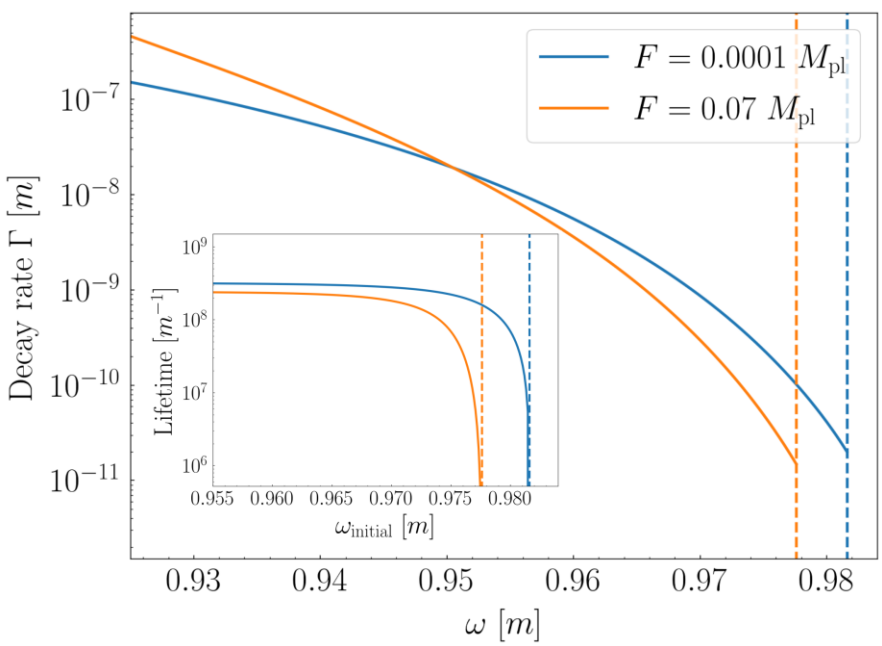
Depending on whether the self-interactions are negligible compared with gravitational interactions, oscillons can be classified into two regimes: dense and dilute ones. We want to explore gravitational effects on lifetimes of dense oscillons.

In the absence of gravity, there are two characteristic features in decay rates of long-lived dense oscillons: (1) Some exceptionally stable intermediate configurations exist (visualized as dips) and (2) The decay rate is dramatically suppressed just before their final collapse at a critical frequency (visualized as downhills). How does gravity affect these properties?

We find that when gravity becomes important, the decay rate in the dip increases and the critical frequency on the downhill decreases. The lifetime of oscillons in both cases is thus *reduced*. In this regime, however, exotic phenomena such as black hole formation might be more interesting than oscillon lifetimes.

$$V(\phi) = \frac{m^2 F^2}{2} \tanh^2 \frac{\phi}{F}$$



$$V(\phi) = m^2 F^2 \left(\sqrt{1 + \frac{\phi^2}{F^2}} - 1 \right)$$